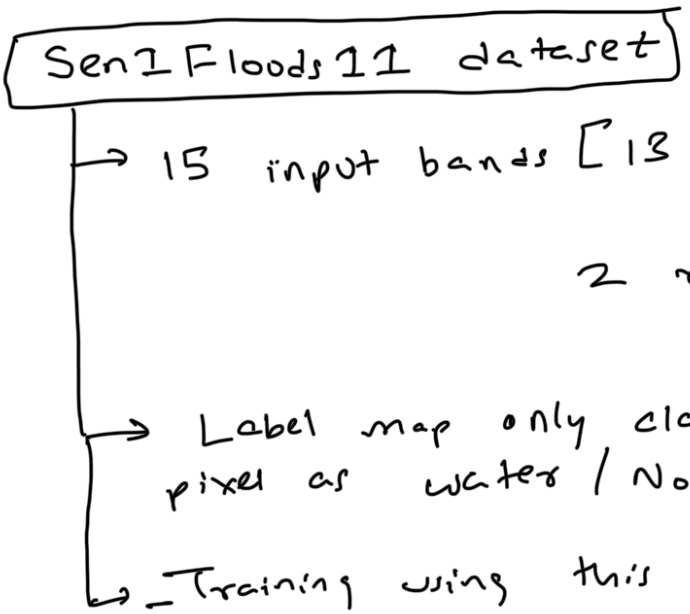
20/10/26

Working on selecting the dataset we need to use for training and finetuning tasks in our project.

Dataset provided by Sidak



The other approach of dataset which can be useful:

Instead of training on a single image of each location during flood time, we create a dataset with multiple images before flood [illegible] flood for all events.

and one during ...
We use all these images to train a model which will predict : land, permanent_water, flood_water and unknown classes. The benefit of this is we directly train a flood inundation model which can predict flood area if we provide satellite images without need of using the differencing approach of before. This can also lead to better compute efficiency as we don't need to run model multiple times to get water map and can get inundation map in a single inference run with model trained on this data.

Doubts that I have:

1) Do we need to use optical image bands for prediction of flood maps as during the flood event, the sky is heavily clouded which can lead to optical input being completely useless for prediction and might lead to errors or failure in prediction?

2) From above if we conclude that optical imagery bands are not useful for flood modelling when sky is not clear [especially the case for India] now we will need to only rely on Sen1 bands data in order to predict inundation map. How would we be able to then deal with limitations of the Sen1 dataset where it can't predict water with enough accuracy and confuses it with lot of other things in the image? What the solution we would need to use for that?

3) Lets consider if we use both (optical + radar) or only radar bands dataset for training of the model and we use the second approach of using both before and during flood event images. Then how and why would the model be able to learn to differentiate between flood water and output them as

and permanent
different classes given that before inaser
are only going to be subsets of the whole
and there is practically going to be no
difference in property of flood and permanent
water ?